

Expt 5 Design of High Pass Filter using Windowing Techniques

5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

Program:

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)

fs = 8000; % Sampling frequency (Hz)

N = 50; % Filter order

wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- 1. FIR High Pass using Rectangular Window ---

b_rect = fir1(N, wc, 'high', rectwin(N+1));

disp('--- High-Pass Coefficients (Rectangular Window) ---');

disp(b_rect);

%% --- 2. FIR High Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'high', hamming(N+1));

disp('--- High-Pass Coefficients (Hamming Window) ---');

disp(b_hamm);

%% --- 3. FIR High Pass using Hanning Window ---

b_hann = fir1(N, wc, 'high', hanning(N+1));

disp('--- High-Pass Coefficients (Hanning Window) ---');

disp(b_hann);

%% --- Frequency Response Plots ---

figure;

freqz(b_rect,1);

title('High-pass FIR using Rectangular Window');

figure;

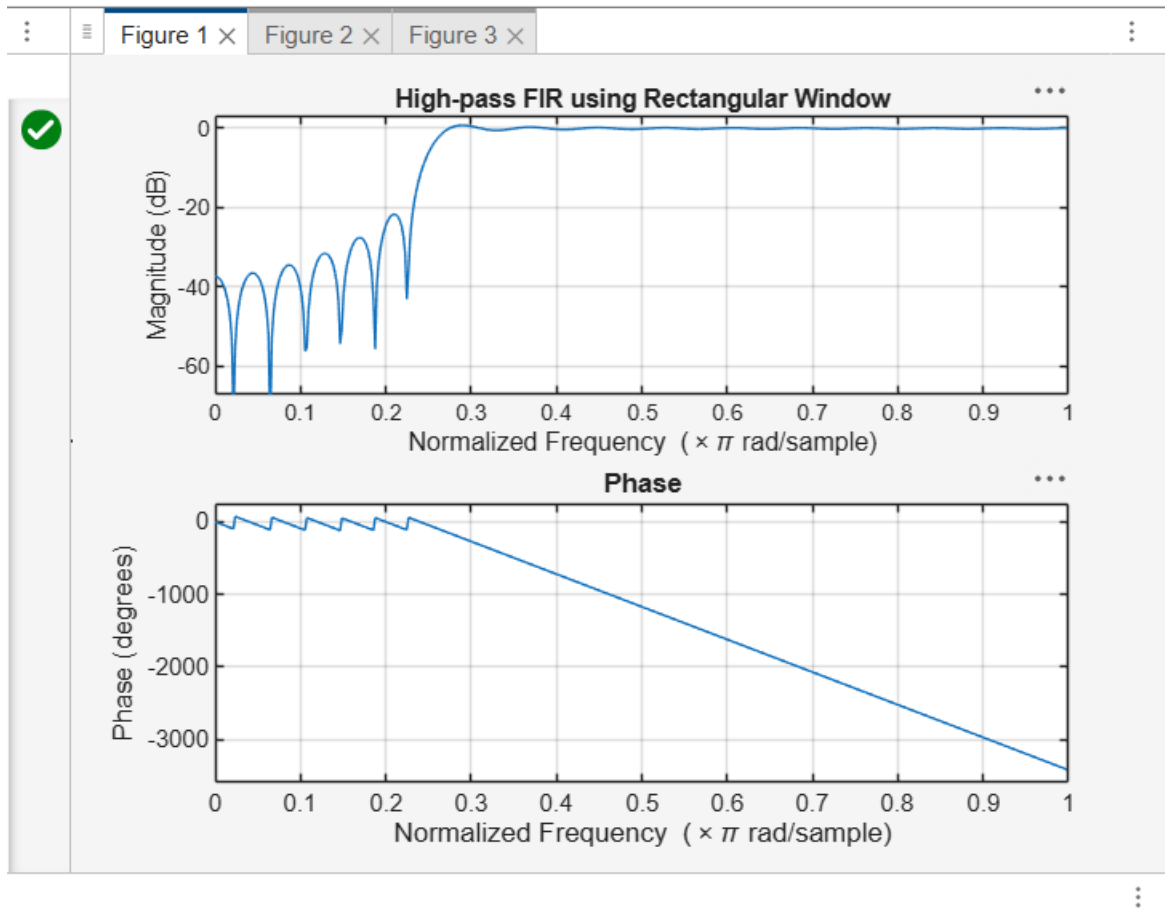
freqz(b_hamm,1);
```

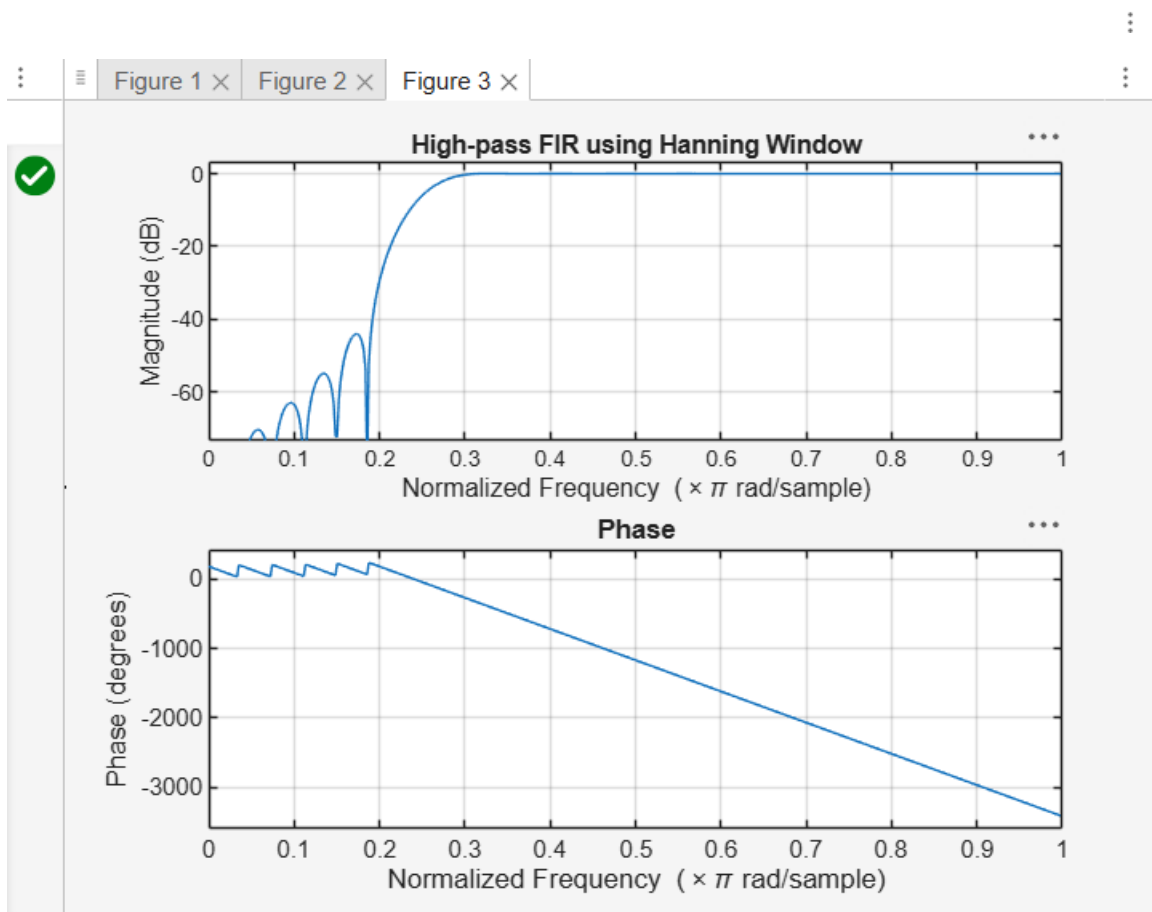
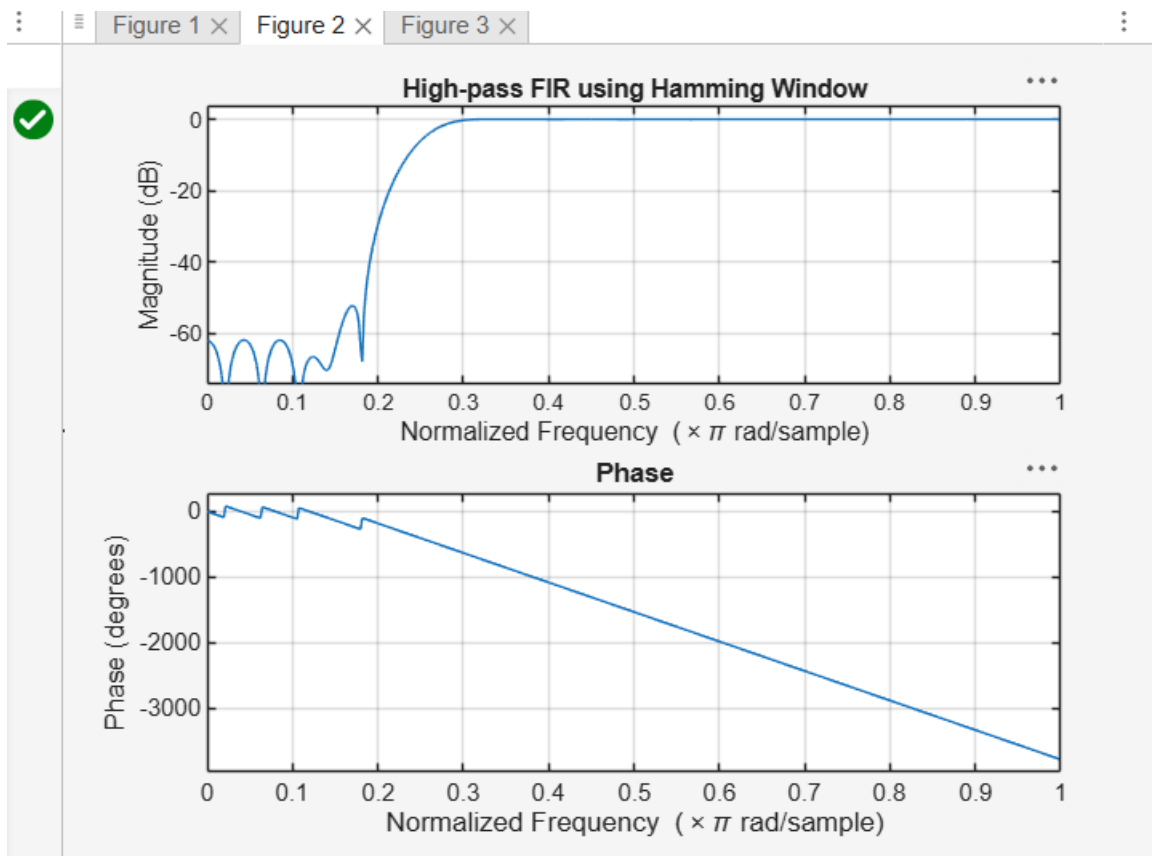
```
title('High-pass FIR using Hamming Window');
```

```
figure;
```

```
freqz(b_hann,1);
```

```
title('High-pass FIR using Hanning Window');
```





5.2. Generate the coefficients of an FIR High-Pass filter using Hamming windowing Techniques

Program:

```
clc; clear; close all;

%% --- Input Parameters ---

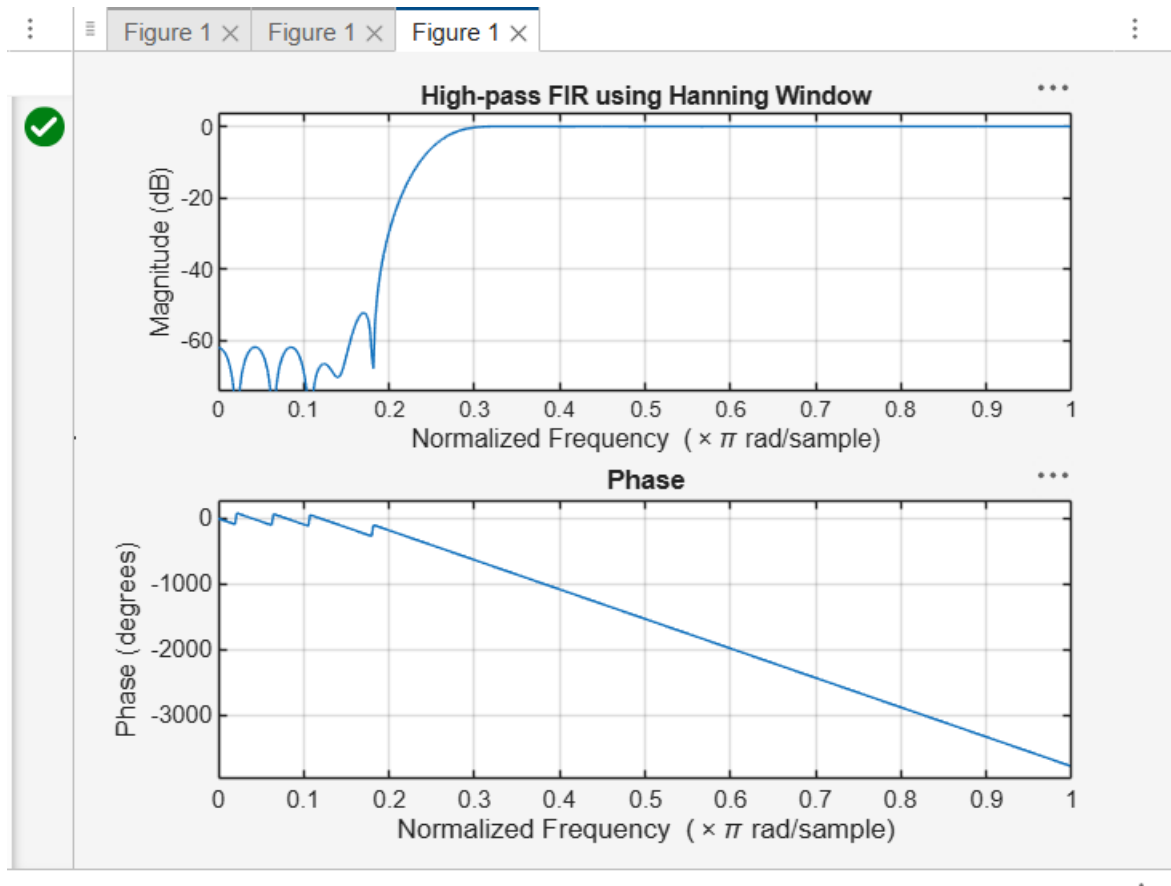
fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%% --- Frequency Response Plots ---

figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');
title('High-pass FIR using Hanning Window');
```



5.3. Generate the coefficients of an FIR High-Pass filter using Hanning windowing Techniques

Program:

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High Pass using Hanning Window ---

b_hann = fir1(N, wc, 'high', hanning(N+1));

disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

%% --- Frequency Response Plots ---

figure;
```

```
freqz(b_hann,1);  
title('High-pass FIR using Hanning Window');
```

